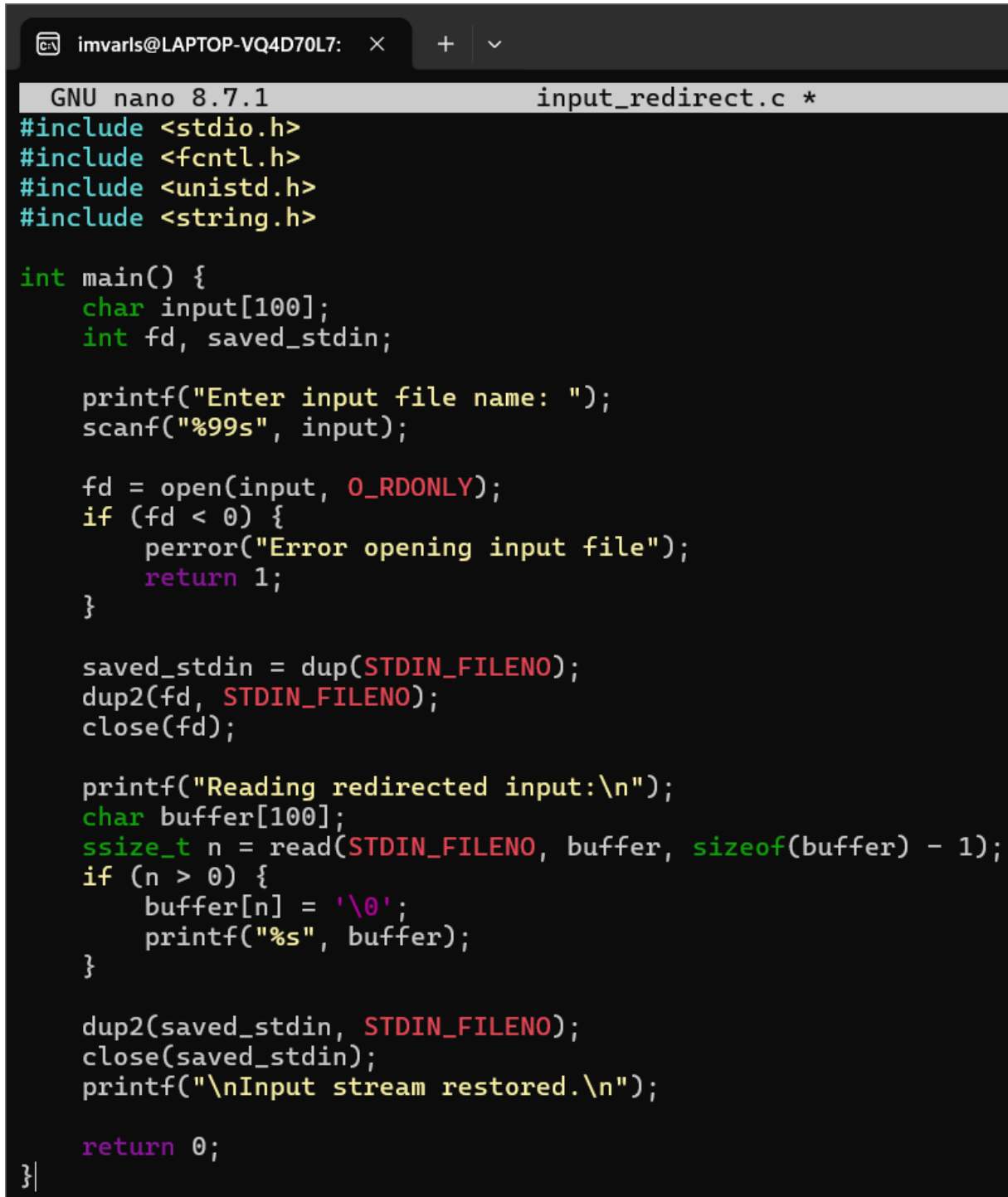


Skill Week - 7

1.To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs



```
GNU nano 8.7.1 input_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    char input[100];
    int fd, saved_stdin;

    printf("Enter input file name: ");
    scanf("%99s", input);

    fd = open(input, O_RDONLY);
    if (fd < 0) {
        perror("Error opening input file");
        return 1;
    }

    saved_stdin = dup(STDIN_FILENO);
    dup2(fd, STDIN_FILENO);
    close(fd);

    printf("Reading redirected input:\n");
    char buffer[100];
    ssize_t n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);
    if (n > 0) {
        buffer[n] = '\0';
        printf("%s", buffer);
    }

    dup2(saved_stdin, STDIN_FILENO);
    close(saved_stdin);
    printf("\nInput stream restored.\n");

    return 0;
}
```

Output:

imvarls@LAPTOP-VQ4D70L7: × + ▾

```
imvarls@LAPTOP-VQ4D70L7:~$ gcc input_redirect.c -o input_redirect
```

```
imvarls@LAPTOP-VQ4D70L7:~$ ./input_redirect
```

```
Enter input file name: input.txt
```

```
Reading redirected input:
```

```
Hello Arun! this is a file copy test.
```

```
Input stream restored.
```

```
imvarls@LAPTOP-VQ4D70L7:~$ anc.txt
```

```
anc.txt: command not found
```

```
imvarls@LAPTOP-VQ4D70L7:~$ ./input_redirect
```

```
Enter input file name: anc.txt
```

```
Error opening input file: No such file or directory
```

```
imvarls@LAPTOP-VQ4D70L7:~$ |
```

2.To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
GNU nano 8.7.1 output_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    char filename[100];
    int fd, saved_stdout;

    printf("Enter output file name: ");
    scanf("%99s", filename);

    fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd < 0) {
        perror("Error opening output file");
        return 1;
    }

    saved_stdout = dup(STDOUT_FILENO);
    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This text is redirected to the output file.\n");
    printf("Output redirection successful.\n");
    fflush(stdout);

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Output stream restored.\n");

    return 0;
}
```

Output:

```
imvarls@LAPTOP-VQ4D70L7: ~$ nano output_redirect.c
imvarls@LAPTOP-VQ4D70L7: ~$ gcc output_redirect.c -o output_redirect
imvarls@LAPTOP-VQ4D70L7: ~$ ./output_redirect
Enter output file name:
output.txt
Output stream restored.
imvarls@LAPTOP-VQ4D70L7: ~$ cat output.txt
This text is redirected to the output file.
Output redirection successful.
imvarls@LAPTOP-VQ4D70L7: ~$
imvarls@LAPTOP-VQ4D70L7: ~$ ls -l output.txt
-rw-r--r-- 1 imvarls imvarls 75 Sep 25 13:44 output.txt
imvarls@LAPTOP-VQ4D70L7: ~$ |
```

3. To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
imvarls@LAPTOP-VQ4D70L7: × + v
GNU nano 8.7.1 append_redirect.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    char filename[100];
    int fd, saved_stdout;
    char *text = "New line appended to the file.\n";

    printf("Enter output file name: ");
    scanf("%99s", filename);

    fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);
    if (fd < 0) {
        perror("Error opening file");
        return 1;
    }

    saved_stdout = dup(STDOUT_FILENO);
    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("%s", text);
    fflush(stdout);

    dup2(saved_stdout, STDOUT_FILENO);
    close(saved_stdout);

    printf("Data appended successfully.\n");

    return 0;
}
```

Output:

imvarls@LAPTOP-VQ4D70L7: ×

+ v

```
imvarls@LAPTOP-VQ4D70L7:~$ nano append_redirect.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc append_redirect.c -o append_redirect
imvarls@LAPTOP-VQ4D70L7:~$ echp "Original data" > append.txt
Command 'echp' not found, did you mean:
  command 'echo' from deb coreutils-from-gnu (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-uutils (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-busybox (0.0.0~ubuntu25)
  command 'echo' from deb coreutils-from-toybox (0.0.0~ubuntu25)
Try: sudo apt install <deb name>
imvarls@LAPTOP-VQ4D70L7:~$ echo "Original data" > append.txt
imvarls@LAPTOP-VQ4D70L7:~$ cat append.txt
Original data
imvarls@LAPTOP-VQ4D70L7:~$ ./append_redirect
Enter output file name: append.txt
Data appended successfully.
imvarls@LAPTOP-VQ4D70L7:~$ cat append.txt
Original data
New line appended to the file.
imvarls@LAPTOP-VQ4D70L7:~$ |
```

4. To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

```
GNU nano 8.7.1 stderr_redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd, saved_stderr;

    fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd < 0) {
        perror("Cannot open error.txt");
        return 1;
    }

    saved_stderr = dup(STDERR_FILENO);
    dup2(fd, STDERR_FILENO);
    close(fd);

    FILE *file = fopen("missing.txt", "r");
    if (file == NULL) {
        perror("Error opening missing.txt");
    } else {
        fclose(file);
    }

    fflush(stderr);

    dup2(saved_stderr, STDERR_FILENO);
    close(saved_stderr);

    printf("stderr restored.\n");

    return 0;
}
```

Output:

```
imvarls@LAPTOP-VQ4D70L7: ~$ nano stderr_redirect.c
imvarls@LAPTOP-VQ4D70L7: ~$ gcc stderr_redirect.c -o stderr_redirect
imvarls@LAPTOP-VQ4D70L7: ~$ ./stderr_redirect
stderr restored.
imvarls@LAPTOP-VQ4D70L7: ~$
imvarls@LAPTOP-VQ4D70L7: ~$ cat error.txt
Error opening missing.txt: No such file or directory
imvarls@LAPTOP-VQ4D70L7: ~$
```